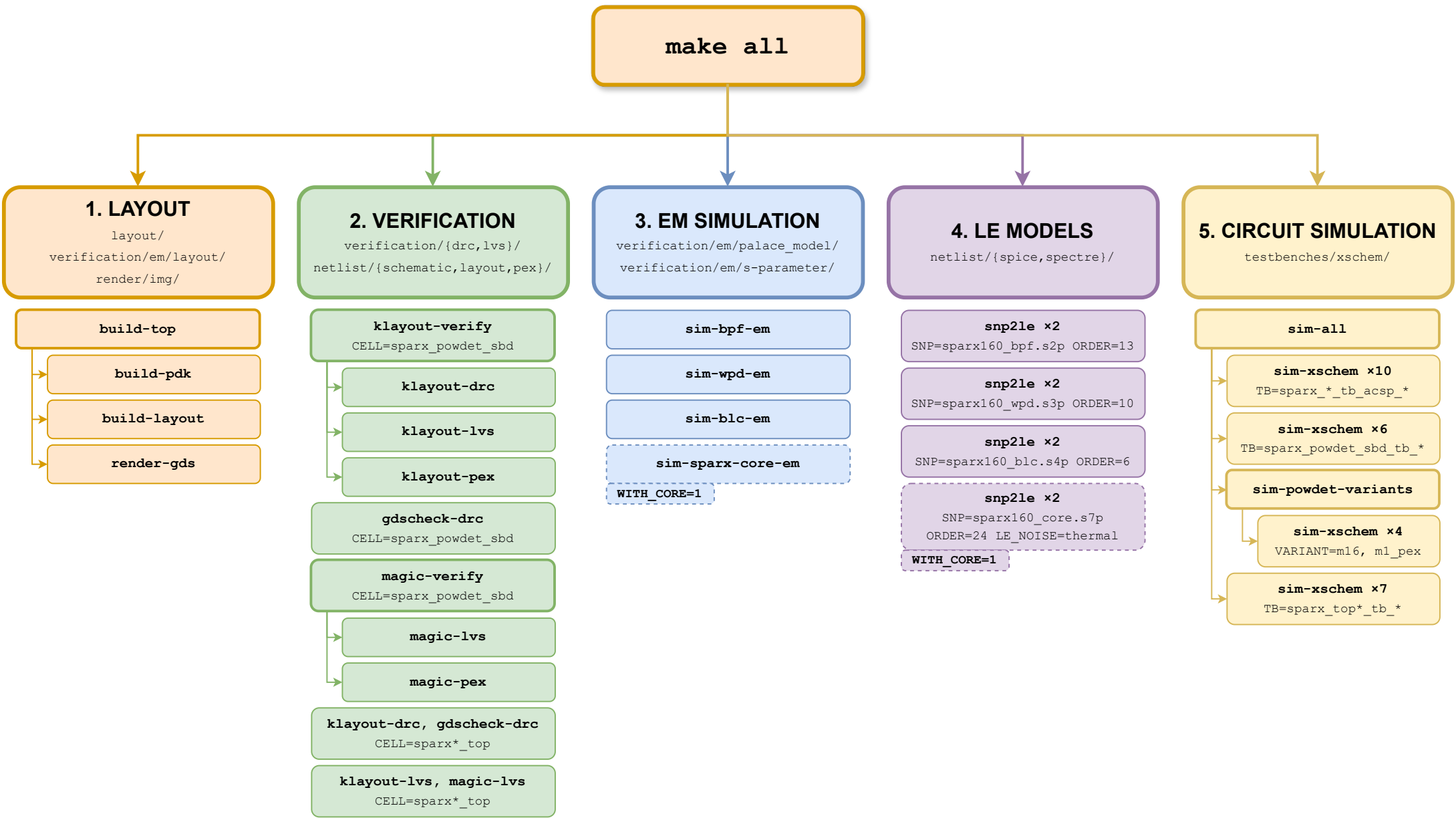

make all
1. LAYOUT
layout/
verification/em/layout/
render/img/
build-top
build-pdk
build-layout
render-gds
2. VERIFICATION
verification/{drc,lvs}/
netlist/{schematic,layout,pex}/
klayout-verify
CELL=sparx_powdet_sbd
klayout-drc
klayout-lvs
klayout-pex
gdscheck-drc
CELL=sparx_powdet_sbd
magic-verify
CELL=sparx_powdet_sbd
magic-lvs
magic-pex
klayout-drc, gdscheck-drc
CELL=sparx*_top
klayout-lvs, magic-lvs
CELL=sparx*_top
3. EM SIMULATION
verification/em/palace_model/
verification/em/s-parameter/
sim-bpf-em
sim-wpd-em
sim-blc-em
sim-sparx-core-em
WITH_CORE=1
4. LE MODELS
netlist/{spice,spectre}/
snp2le ×2
SNP=sparx160_bpf.s2p ORDER=13
snp2le ×2
SNP=sparx160_wpd.s3p ORDER=10
snp2le ×2
SNP=sparx160_blc.s4p ORDER=6
snp2le ×2
SNP=sparx160_core.s7p
ORDER=24 LE_NOISE=thermal
WITH_CORE=1
5. CIRCUIT SIMULATION
testbenches/xschem/
sim-all
sim-xschem ×10
TB=sparx_*_tb_acsp_*
sim-xschem ×6
TB=sparx_powdet_sbd_tb_*
sim-powdet-variants
sim-xschem ×4
VARIANT=m16, m1_pex
sim-xschem ×7
TB=sparx_top*_tb_*